

Basic Rule-Based Chatbot Project

1. Objective

To build a simple rule-based chatbot using Python that interacts with users using predefined responses.

2. Key Concepts Used

- if-elif statements
- Functions
- Loops
- Input/Output
- String Handling

3. Features

- Responds to greetings like 'hello' and 'hi'
- Replies to questions like 'how are you'
- Tells its name
- Responds to 'bye' and exits
- Works continuously until the user exits

4. Python Source Code

```
# -----  
# Simple Rule-Based Chatbot  
# -----  
  
def chatbot():  
    print("=====  
    print(" SIMPLE CHATBOT PROJECT ")  
    print("=====  
    print("Type 'bye' to exit the chatbot.")  
  
    while True:  
        user_input = input("You: ").lower()  
  
        if user_input == "hello" or user_input == "hi":  
            print("Bot: Hi! How can I help you?")  
  
        elif user_input == "how are you":  
            print("Bot: I'm fine, thanks!")  
  
        elif user_input == "what is your name":  
            print("Bot: My name is ChatBot.")  
  
        elif user_input == "help":
```

```
        print("Bot: I can chat with you. Try saying hello!")

    elif user_input == "bye":
        print("Bot: Goodbye! Have a nice day.")
        break

    else:
        print("Bot: Sorry, I don't understand that.")

chatbot()
```

5. Sample Output

```
You: hello
Bot: Hi! How can I help you?

You: how are you
Bot: I'm fine, thanks!

You: what is your name
Bot: My name is ChatBot.

You: bye
Bot: Goodbye! Have a nice day.
```

6. Explanation of the Program

- The chatbot uses a function named chatbot().
- while True creates an infinite loop for continuous chatting.
- input() takes user messages.
- .lower() converts text into lowercase.
- if-elif conditions check user messages and generate replies.
- break statement exits the chatbot when the user types 'bye'.

7. Advantages

- Easy to understand and implement
- Beginner-friendly project
- Improves Python programming skills
- Demonstrates chatbot working principles

8. Limitations

- Cannot understand complex sentences
- Uses only predefined responses
- No AI or Machine Learning capability

9. Future Improvements

- Add more conversational responses
- Add Artificial Intelligence and NLP
- Create GUI using Tkinter
- Add voice recognition support

10. Conclusion

This project demonstrates the implementation of a simple rule-based chatbot using Python. The chatbot interacts with users using predefined rules and helps beginners understand basic chatbot development concepts.